

Derivation of Cochran's Formula for Sample Size Determination

1 The Central Limit Theorem: Foundation of Statistical Inference

The Central Limit Theorem (CLT) is the cornerstone of statistical inference and serves as the theoretical foundation for Cochran's formula. The theorem states:

Suppose $X_1, X_2, X_3, \dots$ is a sequence of independent and identically distributed random variables with:

- Expected value $E[X_i] = \mu$
- Variance $\text{Var}[X_i] = \sigma^2 < \infty$

Then, as the sample size m approaches infinity, the standardized sample mean converges in distribution to a standard normal distribution:

$$\sqrt{m}(\bar{X}_m - \mu) \xrightarrow{d} N(0, \sigma^2) \quad (1)$$

The CLT requires two critical conditions:

- Finite mean (ensuring the existence of a central value to converge to)
- Bounded variability

2 Properties of the Normal Distribution and Critical Values

For a standard normal distribution $Z \sim N(0, 1)$, we define $z_{\alpha/2}$ as the critical value where:

- α is our significance level (for 95% confidence, $\alpha = 0.05$)
- $z_{\alpha/2}$ is the value such that $P(Z > z_{\alpha/2}) = \alpha/2$

For example, with 95% confidence ($\alpha = 0.05$):

- $z_{0.025} = 1.96$
- $P(Z > 1.96) = 0.025$

The probability that Z falls between $-z_{\alpha/2}$ and $z_{\alpha/2}$ is:

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha \quad (2)$$

Figure 1 illustrates these concepts visually, showing how the critical values $z_{\alpha/2} = \pm 1.96$ divide the standard normal distribution into confidence and rejection regions for a 95% confidence level.

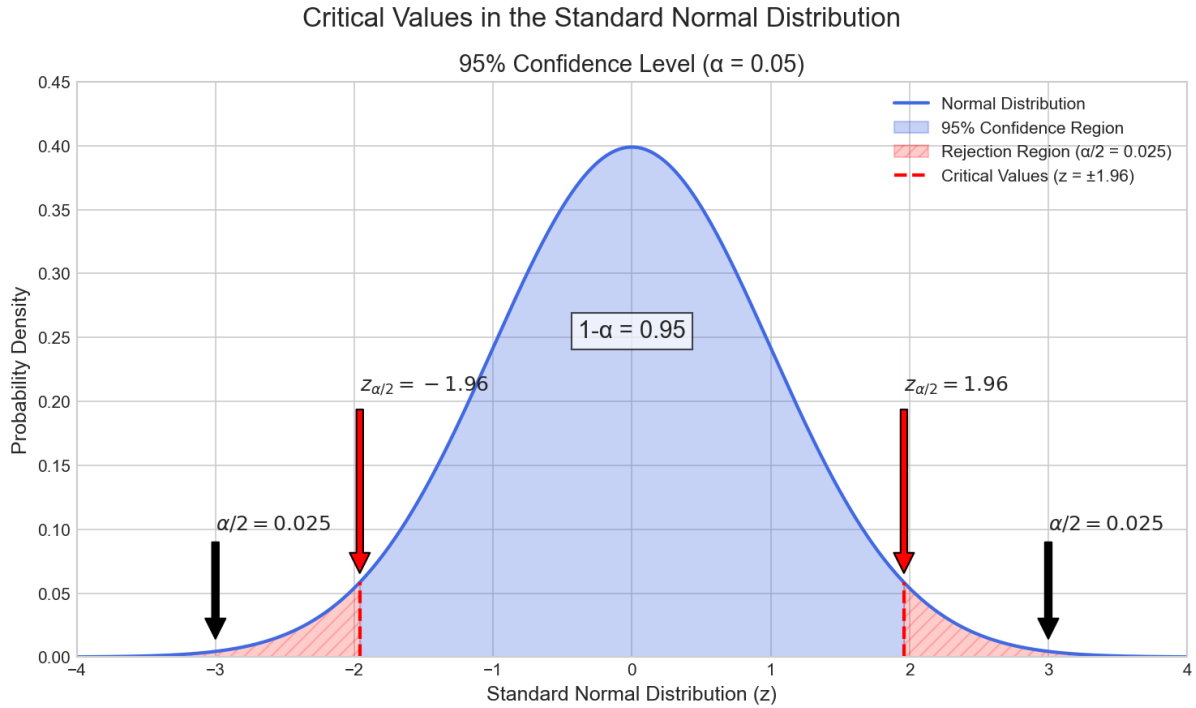


Figure 1: Visualization of critical values for a 95% confidence level in the standard normal distribution.

As shown in the figure, the critical values serve as boundary points that separate the 95% confidence region from the rejection regions. When constructing confidence intervals, these critical values are essential for determining the margin of error and ensuring the desired level of confidence.

3 From Sample Mean to Confidence Intervals

When we have a sample mean $\bar{X}$ from a population with mean μ and standard deviation σ , the CLT tells us that the standardized sample mean follows a normal distribution:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{m}} \sim N(0, 1) \quad (3)$$

This allows us to make probability statements about how far our sample mean might deviate from the true population mean:

$$P\left(-z_{\alpha/2} < \frac{\bar{X} - \mu}{\sigma/\sqrt{m}} < z_{\alpha/2}\right) = 1 - \alpha \quad (4)$$

Rearranging to isolate μ :

$$P\left(\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{m}} < \mu < \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{m}}\right) = 1 - \alpha \quad (5)$$

4 The Margin of Error

The margin of error E represents half the width of the confidence interval:

$$E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{m}} \quad (6)$$

This formula shows that the margin of error:

- Increases with the confidence level (larger $z_{\alpha/2}$)
- Increases with population variability (larger σ)
- Decreases with sample size (larger m)

5 Cochran's Key Insight: Flipping the Perspective

Cochran's formula represents a shift in perspective. Instead of calculating a confidence interval width after sampling, we decide beforehand how precise we want our estimate to be:

1. We set a desired margin of error E
2. We solve for the required sample size m

Rearranging the margin of error formula:

$$E = z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{m}} \quad (7)$$

Squaring both sides:

$$E^2 = (z_{\alpha/2})^2 \cdot \frac{\sigma^2}{m} \quad (8)$$

Solving for m :

$$m = \frac{(z_{\alpha/2})^2 \cdot \sigma^2}{E^2} \quad (9)$$

This is Cochran's formula for determining the required sample size when:

- We know the population variance σ^2
- We specify a confidence level (determines $z_{\alpha/2}$)
- We specify a desired margin of error E

In practice, the population variance σ^2 is often unknown and must be estimated. This can be done using the sample variance formula:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} \quad (10)$$

Where x_i are the individual observations, $\bar{x}$ is the sample mean, and n is the number of observations in the sample.

6 Cochran's Formula for Estimating Proportions

A particularly important application of Cochran's formula is for estimating population proportions, which follows directly from understanding Bernoulli random variables:

6.1 Bernoulli Random Variables

Bernoulli random variables represent binary outcomes:

- Each observation is either a success (1) with probability p or a failure (0) with probability $1 - p$
- A single Bernoulli trial X follows a Bernoulli distribution with parameter p
- $P(X = 1) = p, P(X = 0) = 1 - p$
- $E(X) = p, \text{Var}(X) = p(1 - p)$

6.2 Binomial Distribution for Multiple Trials

The sum of m independent Bernoulli trials $Y = X_1 + X_2 + \dots + X_m$ follows a binomial distribution with parameters m and p :

- $Y \sim \text{Bin}(m, p)$
- $E(Y) = mp$
- $\text{Var}(Y) = mp(1 - p)$

6.3 Sample Proportion and its Properties

When estimating a proportion:

- Each individual observation is a Bernoulli trial
- The total number of "successes" follows a binomial distribution
- The sample proportion $\hat{p} = \frac{Y}{m}$ is the binomial count divided by the number of trials

Properties of the sample proportion:

- Mean: $E[\hat{p}] = p$
- Variance: $\text{Var}(\hat{p}) = \frac{p(1-p)}{m}$

6.4 Applying the Central Limit Theorem to Proportions

For sufficiently large m , the CLT tells us that $\hat{p}$ is approximately normally distributed:

- $\hat{p} \sim N(p, \sigma^2)$, where $\sigma^2 = \frac{p(1-p)}{m}$
- The standardized form: $\frac{\hat{p} - p}{\sqrt{p(1-p)/m}} \sim N(0, 1)$

6.5 Confidence Interval for Proportions

$$P\left(-z_{\alpha/2} < \frac{\hat{p} - p}{\sqrt{p(1-p)/m}} < z_{\alpha/2}\right) = 1 - \alpha \quad (11)$$

Rearranging to isolate p :

$$P\left(\hat{p} - z_{\alpha/2} \cdot \sqrt{\frac{p(1-p)}{m}} < p < \hat{p} + z_{\alpha/2} \cdot \sqrt{\frac{p(1-p)}{m}}\right) = 1 - \alpha \quad (12)$$

6.6 Margin of Error for Proportions

The margin of error is:

$$E = z_{\alpha/2} \cdot \sqrt{\frac{p(1-p)}{m}} \quad (13)$$

6.7 Cochran's Formula for Proportions

Solving for the required sample size:

$$m = \frac{p(1-p)(z_{\alpha/2})^2}{E^2} \quad (14)$$

This is Cochran's formula specifically for estimating proportions, showing that the required sample size depends on:

- The confidence level (determines $z_{\alpha/2}$)
- The margin of error E we're willing to accept
- The proportion p (or our best guess of it)

When p is unknown, we often use $p = 0.5$, which maximizes $p(1-p)$ and gives us the most conservative (largest) sample size.